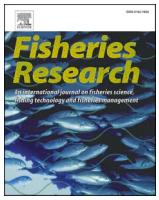
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Large males matter: Low sperm reserves in female snow crab (*Chionoecetes opilio*) off Newfoundland, Canada

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ABSTRACT

Large, male snow crab (*Chionoecetes opilio*) off Newfoundland and Labrador were heavily exploited throughout their range from 2015 to 2018 leading to biological concerns about the possibility of sperm limitation within portions of the stock. To address this, primiparous females were sampled from six sites off the island of Newfoundland to gather data on their overall reproductive health (e.g., seminal receptacle load, SL, and clutch weight) at the end of this heavily exploited period. Snow crab females had low SLs in comparison to long-term monitoring at nearby sites in the Northern Gulf of St. Lawrence but were similar to historic observations from the Eastern Bering Sea. Seminal receptacle loads were significantly influenced by the female's carapace width, time since mating (represented by sampling month), as well as site-specific characteristics of the male population (ratio of most competitive males in the population and male time since terminal molt). Clutch weight was significantly influenced by the female's carapace width, as well as the average size of sexually mature males and the average time since terminal molt of adult males. Our findings show the potential for sperm limitation to occur in populations depleted of large-male snow crab and by extension for excessive fishing of males to diminish reproductive capacity of the stock. These findings highlight the importance of maintaining conservative exploitation rates for snow crab that protect a substantial proportion of large adult males in order to preserve reproductive capacity.

1. Introduction

Snow crab (*Chionoecetes opilio*) is a stenothermic species in sub-arctic and arctic waters and is targeted by commercial fisheries throughout much of its range. Snow crab exhibit sexual size dimorphism, with adult males normally significantly larger than adult females, leading to male-only harvest in most regions where this species is fished. Size-related gear restrictions and female and sublegal male discard regulations generally allow for all females and a portion of reproductive males to escape fishing mortality (Watson, 1970; Kruse, 1993). As such, snow crab fisheries targeting only large males are thought to be inherently precautious (Ennis et al., 1990; Kruse, 1993) and despite the reproductive biology of snow crab being well understood (Elner and Beninger, 1992, 1995; Sainte-Marie et al., 2008), relatively little attention has been paid to monitoring the overall reproductive health of the stocks.

In the Northwest Atlantic, female snow crab typically reach sexual maturity at 6–8 years of age (Alunno-Bruscia and Sainte-Marie, 1998; Comeau et al., 1998a). At this time, they terminally molt (i.e., become adult) and develop physical features to facilitate egg brooding (Watson, 1970). Prior to this molt, female snow crab have a pubescent stage where chemical cues entice males to begin pre-copulatory guarding (Bouchard et al., 1996; Bublitz et al., 2008), although insemination can only occur after the female terminally molts (Sainte-Marie et al., 1999). During this first mating, females are inseminated while in a soft-shell condition, and are vulnerable to mortality from competing males or predation. Accordingly, male mate-guarding is critically important (Sainte-Marie et al., 2008). Depending on the socio-sexual context, males may choose to guard females prior to, during, and/or after her terminal molt to ensure paternity (Rondeau and Sainte-Marie, 2001; Sainte-Marie et al., 2008).

A mature female snow crab carrying eggs during her first breeding season is referred to as primiparous (Sainte-Marie et al., 2008). Pubescent-primiparous females may be mated by more than one male and have been documented storing the sperm of up to six males within their seminal receptacle, designated as spermatheca in former snow crab literature (Urbani et al., 1998; Sainte-Marie et al., 1999) (from hereon

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primiparous will be used to describe pubescent and recently matured primiparous females). After one breeding season, egg-bearing females are termed multiparous (Sainte-Marie et al., 2008). Multiparous females may use sperm stored in their seminal receptacles from the previous mating season to fertilize an egg clutch or mate again (Watson, 1970; Elner and Beninger, 1992).

Primiparous and multiparous female snow crab tend to aggregate separately, have different breeding seasons, and may be mated by different groups of males (Ernst et al., 2005; Sainte-Marie et al., 2008; Nichol and Somerton, 2015; Murphy et al., 2017). In eastern Canada, primiparous females often form loose aggregations in shallow waters, where individuals mature asynchronously during the winter months (Alunno-Bruscia and Sainte-Marie, 1998; Sainte-Marie et al., 1999). Multiparous females commonly form dense aggregations in deeper waters, and release larvae that often coincide with spring blooms (Ennis et al., 1990; Starr et al., 1994; Sainte-Marie et al., 2008; Kuhn et al., 2011). At any given carapace width (CW), primiparous females of the genus *Chionoecetes* are less fecund than multiparous females (Somerton and Meyers, 1983) but may produce larger, more energy-rich eggs (Sainte-Marie, 1993; Sainte-Marie et al., 2008).

The likelihood of successful mating and reproductive potential for an individual male snow crab are a function of their overall body size, size of chelae, shell hardness, age, and spatial distribution (Sainte-Marie et al., 1999, 2008; Slater et al., 2010). Males sexually mature before terminally molting, during which time they are termed adolescent (Sainte-Marie et al., 1995). However, features that increase male sexual competitiveness, such as longer limbs and enlarged chelae used for grasping females during mating (Elner and Beninger, 1992), do not form until their terminal molt into adulthood (Conan and Comeau, 1986; Stevens et al., 1993; Chabot et al., 2008). Adult males are more competitive in the mating process than adolescents (Claxton et al., 1994) and, in general, larger adults are more successful than smaller adults at competing for mating opportunities (Sainte-Marie et al., 1997; Comeau et al., 1998b). Sainte-Marie et al. (1999) found that adult males paired with sexually receptive primiparous females were larger and had harder shells and fewer missing limbs compared to adult males in non-fecund pairs or to males in the overall population. Males can survive for 7–8 years after terminal molt (Fonseca et al., 2008), but breeding success may peak at 2–5 years post-terminal molt, with accruing injuries and senescence diminishing reproductive fitness in later years (Sainte-Marie et al., 1995; Fonseca et al., 2008).

Larger male snow crab have a greater number of spermatophores within their gonads compared to smaller males (Fedoseyev, 1988), and males with larger gonads will ejaculate larger sperm loads than males with smaller gonads (Rondeau and Sainte-Marie, 2001). In addition, compared to adolescent males, adult males pass larger volumes of seminal fluid that can displace previously deposited ejaculates away from the oviduct and intermediate chamber of the seminal receptacle (Sainte-Marie and Lovrich, 1994), thereby ensuring priority usage of their sperm (Urbani et al., 1998).

Male snow crab have flexible mating behaviors and can adapt these based on perceived competition (Sainte-Marie et al., 2008). For example, male snow crab adjust their guarding behavior in relation to the sociosexual context: in laboratory experiments, Rondeau and Sainte-Marie (2001) demonstrated that as the sex ratio increased in favor of males, male snow crab guarded longer and females had higher rates of mortality and injury as a result of competition and takeover attempts by rival males. Male snow crab also practice "sperm economy" behavior. Males do not routinely adjust their size of ejaculation to the size of the female (Adams and Paul, 1983; Sainte-Marie and Lovrich, 1994), but instead scale the ejaculation strategically in relation to perceived risk of sperm competition and number of mating opportunities (Rondeau and Sainte-Marie, 2001). Furthermore, in crustaceans, if the sex ratio is skewed toward females, males may mate more often, but as the frequency of matings increases, the size of the ejaculation will be reduced (Sato et al., 2005; Sato, 2011). This sperm economization with increasing frequency of mating reflects depletion of individual sperm reserves, with depleted snow crab males allocating less sperm than less depleted males with the same number of mating opportunities (Rondeau and Sainte-Marie, 2001).

Sperm economy and sperm depletion create the opportunity for sperm limitation to occur as a result of relaxed intraspecific competition in response to strong fishing pressure in male-only fisheries (Carver et al., 2005; Sainte-Marie et al., 2008; Sato, 2011). Sperm limitation occurs when there are not enough sperm (quality and/or quantity) to sufficiently fertilize all eggs produced by females in the population. This can occur in a variety of taxa when there are not enough males to inseminate all mature females, there is insufficient time to replenish sperm reserves between matings, or when males deliver too few sperm in order to mate multiple females (e.g., Warner et al., 1995; MacDiarmid and Butler, 1999). A fishery targeting only large males could impact the overall snow crab population in a variety of ways, including reduced availability of adult (versus adolescent) males, reduced adult sex ratio (males to total adults), and reduced average male size (Mullowney and Baker, 2020a), all of which have the potential to cause sperm limitation (Sainte-Marie et al., 2008; Sato, 2011; Pardo et al., 2017).

The issue of fishing-induced sperm limitation has been identified as a concern for some brachyuran crabs. In the Chilean rock crab (*Metacarcinus edwardsii*), males were more sperm-depleted and females had lower sperm reserves and brood fecundity in areas where the intensity of male, size-selective harvest was high, compared to areas where the harvest intensity was low (Pardo et al., 2015, 2017). In Chesapeake Bay (USA), researchers have raised concerns over the potential for fishing-induced sperm limitation in blue crab (*Callinectes sapidus*). For example, Carver et al. (2005) found that some male blue crab in areas subjected to higher fishing pressure were sperm depleted and those that were most depleted did not even attempt to mate. Ogburn et al. (2019) found that spawning stock reproductive output declined by 5–10% as a result of sperm limitation in female blue crabs mainly during their second spawning season. Although snow crab have been harvested throughout the world for many years, evidence of broad-scale, fishing-induced sperm limitation has not been documented in wild populations of this species.

The Newfoundland and Labrador (NL) snow crab fishery represents one of the most valuable fisheries in Canada, with landings in 2021 valued at over $600 million (DFO, unpublished data). The fishery targets only the most sexually competitive males (hard-shelled, > 94 mm CW) using large-mesh (~140 mm) pots and is controlled through quotas, seasons, limited access, and gear and trip restrictions (DFO, 2019).

The NL snow crab exploitable biomass declined substantially during 2015–2018 to historic low levels without the total harvest being reduced in an equivalent proportion, leading to exceptionally high exploitation levels in some areas. On the broad-scale, virtually all portions of the approximately 200,000 nm^2 stock range experienced exploitation rates above 50% of the estimated exploitable biomass in one or more years during the 2015–2018 period, with some areas reaching levels above 75% annual exploitation. Indications of overexploitation up until 2018 became evident across much of the stock range (Mullowney and Baker, 2020a, 2020b; Mullowney et al., 2020; Baker et al., 2021); male size at terminal molt significantly declined in multiple areas (Mullowney and Baker, 2020a), discard levels of soft-shell male crab reached levels greater than 20% in most areas causing an unknown amount of additional (indirect) fishing mortality (Mullowney et al., 2020), and several sites started to exhibit declining trends in the percentage of females with full egg clutches (Baker et al., 2021).

The recent series of events surrounding unusually high levels of exploitation in the NL snow crab stock effectively created an in-situ experiment on potential for sperm limitation to occur in wild male-only snow crab fisheries. The combination of observations including high exploitation rates, changes in size at terminal molt, and possible trends in dampened egg clutch fullness witnessed throughout much of the expansive stock range led us to be concerned about the possibility of

sperm limitation in the NL snow crab stock in recent years. Specifically within this paper, we investigate seminal receptacle load (SL) levels, identify factors influencing SL, and evaluate if and how these factors impact clutch size in primiparous female snow crab off NL.

2. Materials and methods

2.1. Site description

Since all potentially productive snow crab areas off NL have been subjected to fishing pressure for over 25 years, no control site (i.e., area free of fishing) could be identified. As such, a variety of sites were identified that covered a wide geographical range, were accessible for sampling, and were thought to represent a possible range of contrasting fishing pressures and snow crab population metrics (Fig. 1) (Baker et al., 2021). It should be noted that ontogenetic and seasonal movements are thought to be relatively large (25–70 km) in many portions of the stock range (Mullowney et al., 2018a), thus exploitation rates and breeding dynamics are likely subject to broader than localized-scale processes in most cases. The NL snow crab stock assessment quantifies fishery exploitation rates for management purposes at spatial scales larger than the study sites examined herein (see Baker et al., 2021), thus estimates of localized exploitation rates are unavailable and instead are viewed as relative levels and trends through the available time series.

Six study sites off the island of Newfoundland were chosen for sampling. White Bay is off the northeast coast of the island of Newfoundland and represents the most northerly site chosen for this study (Fig. 1). It is a relatively deep (400–500 m) fjord protected at the mouth by a shallower sill (200–300 m). Notre Dame Bay is a wide bay, also found on the northeast coast of Newfoundland, featuring relatively shallow nearshore waters that connect to deeper (300–400 m) trenches. Exploitable biomass indices in both White Bay and Notre Dame Bay, as well as in adjacent areas dropped steadily and steeply from 2013 to 2018, with exploitation rate indices progressively increasing during this time (Baker et al., 2021). In 2017 and 2018, exploitation rates in these two bays reached near time-series highs (Baker et al., 2021). White Bay showed a significant drop in fishery catch per unit effort from 2013 to 2018, while Notre Dame Bay had steady, oscillating fishery catch rates (Baker et al., 2021).

Conception Bay is located within the Avalon Peninsula of Newfoundland and consists of moderately deep (200–400 m) water in its central interior and a shallow sill near its mouth (100–200 m). Conception Bay was long-been characterized as productive snow crab habitat, but consecutive years of little to no recruitment caused this area to experience a drastic decline in the exploitable biomass index from 2014 to 2018 and a steadily increasing trend in the exploitation rate index associated with non-prohibitive quotas (Baker et al., 2021). Beginning in 2016, there was an unprecedented abrupt decline in fishery catch rates and at the time of sampling for this study (2018) the area was considered to have very low snow crab productivity (Baker et al., 2021).

3L200 is an offshore area that extends past the slope edge of the continental shelf, where snow crab can be found in waters greater than 300 m. The naming convention of this area represents Northwestern Atlantic Fisheries Organization (NAFO) Division 3L, beyond the economic exclusive zone (i.e., 200-mile limit). Consistent with most areas of the stock range, from 2014 to 2018 this area experienced a significant and drastic decline in the exploitable biomass index (Baker et al., 2021). In 2018, exploitation rates reached a 10-year high, but remained lower than the time-series high recorded in 2006. In recent years, both recruits and older hard-shell male crab were relatively low in numbers and fishery catch rates declined to low levels in response to the resource decline.

Placentia Bay and Fortune Bay are located off the south coast of Newfoundland. Placentia Bay is a large bay that forms a wide continuum

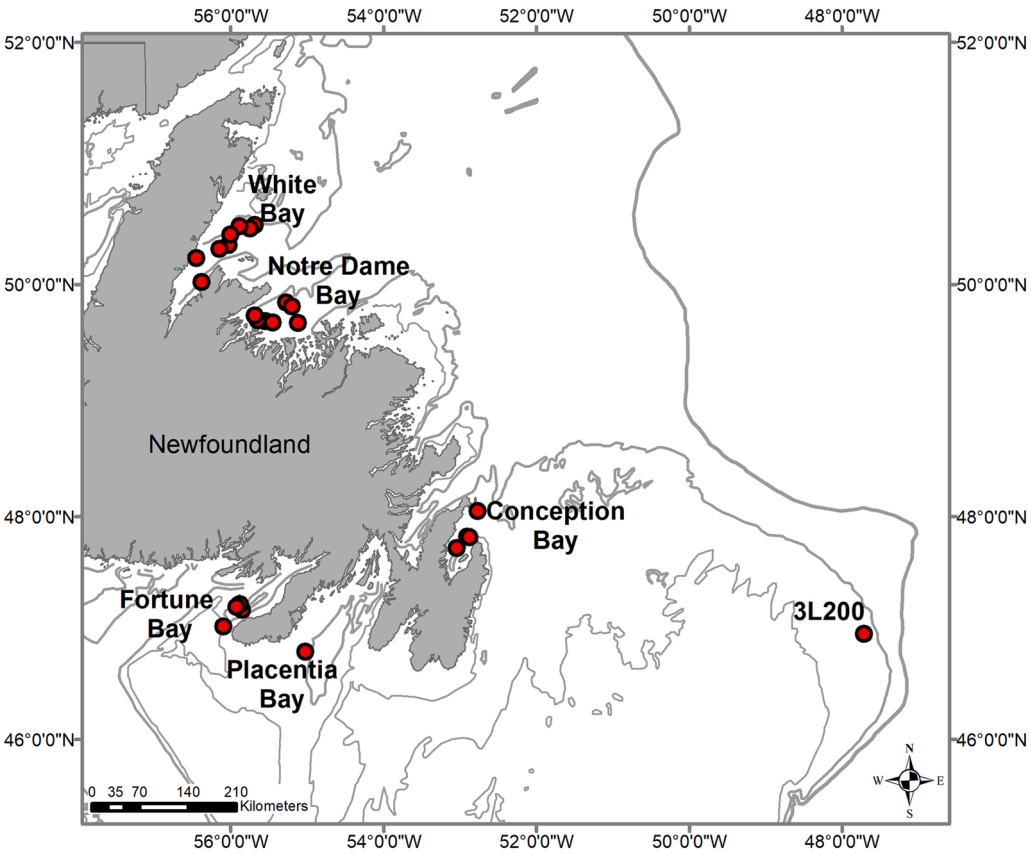


Fig. 1. Sets where female snow crab were sampled in 2018 off Newfoundland. Site names used throughout the analyses are identified.

with the offshore, particularly in context of the spatially expansive Halibut Channel that extends north-south along virtually the entire length of Placentia Bay to the offshore shelf break. Between 2008 and 2017, the exploitable biomass index gradually declined to historical lows, but a large recruitment pulse occurred in 2018. The exploitation rate index for Placentia Bay, the dominant bay and fishery of the Newfoundland south coast, is unknown in 2017 due to poor survey coverage. The exploitation rate peaked in 2014, but fell drastically between 2014 and 2016 as a result of reductions in quota and catches.

Fortune Bay is a relatively discrete bay situated off the central portion of southern Newfoundland. Shallow peripheries of less than 100 m depth surround the deeper central confines of the bay, which can exceed 300 m depth, on all sides (Baker et al., 2021). In 2014, the exploitable biomass index started to show a significant decline, however, large quota cuts were implemented and the exploitation rate indices declined in turn (Baker et al., 2021). Fortune Bay has only experienced short and infrequent periods of productive snow crab fisheries over the past three decades and alternated between phases of commercial versus monitoring fisheries. Since 2014, quotas have been kept relatively small in Fortune Bay to maintain exploitation rates with the lowest values in the time series. However, survey and fishery catch rates remained low during the study period, as they have for most of the past two decades (Baker et al., 2021). In contrast to all other study sites, the prolonged poor state of snow crab within Fortune Bay creates potential to examine if persistent low density and unproductive conditions create differential responses in snow crab reproduction dynamics.

2.2. Collection of samples

A total of 30–60 berried primiparous females were collected opportunistically at each site using small-mesh (27 mm) crab pots during dedicated Fisheries and Oceans Canada (DFO) inshore monitoring surveys, post-season industry-DFO collaborative surveys, alternative research projects, and fishing operations in 2018. Primiparous females were identified based on their soft to relatively hard and clean exoskeleton, iridescent chelae (dorsal view), pinkish to creamy white and broad abdominal flap, and dactyl tips with little to no sign of wear. Females collected during DFO inshore surveys were sampled onboard the vessel, while all other females were transported alive to St. John's, NL and sampled as soon as possible.

Carapace width (CW_F), abdominal width measured across the fifth abdominal segment (AW_F), shell condition, egg stage, number of missing limbs (prior to losses as a result of survey operations, as indicated by the age of scars), and ovary size and color were recorded for all females. Shell condition was categorized into one of two options: (1) clean and soft, (2) clean and hard. Clutches were subjectively categorized as full or partial.

Both seminal receptacles were removed. The right seminal receptacle was placed in 4% formalin and the left in 99% ethanol. Egg clutches (with pleopod endopods) were also placed in 99% ethanol. The fixative was replaced within 1–2 weeks if laboratory analyses could not be completed before this time.

2.3. Laboratory methodology

Egg clutches with endopods were weighed to the nearest 0.1 mg. The content of the right seminal receptacle was isolated by peeling away surrounding maternal tissue and the SL was determined to the nearest 0.0001 g. Sperm counts (millions of sperm cells) were completed on a subset of the right seminal receptacle to ensure SL could be used as a proxy for sperm count (i.e., that there was a strong relationship between the two measurements). At each site, 9–14 right seminal receptacle contents were selected for sperm count to evenly cover the full range of local SLs. Seminal receptacle content was homogenized and diluted to a known volume with water and a small amount of methylene blue (see detailed methods in Sainte-Marie and Lovrich, 1994). Three replicate hemacytometer samples were drawn from the homogenate, sperm were counted under a compound microscope, and the mean count was extrapolated to homogenate volume to estimate the total number of sperm in the seminal receptacle.

2.4. Survey data

Biometric variables that could influence SL and clutch size were calculated for each of the six study sites. These were derived from data collected during annual DFO inshore trap surveys or post-season industry-DFO collaborative trap surveys. All surveys occur in the fall (late August to November), except Fortune Bay which occurs in-season during the late spring (late-May to June). Details regarding these surveys can be found in Baker et al. (2021). Both suites of surveys use similar data collection methodologies (e.g., data recorded and trap size), but DFO inshore trap surveys have more repeated sampling at each set (Baker et al., 2021) and within the site as a whole. At any given sampling station three of each mesh size pot (27-mm, small-mesh pots and 140-mm, large-mesh pots) are sampled in the DFO surveys, and only one of each is sampled during the post-season industry-DFO collaborative surveys. The CW, chela height, shell condition (with categories of very soft, soft, new hard, intermediate, old, or very old), and weight of male snow crab are recorded. Carapace width, shell condition, maturity, and clutch fullness are recorded for every female snow crab (Baker et al., 2021).

To ensure representative captures of both females and small males, data from small-mesh pots were used for analyses. Since adolescent (non-terminally molted) males have the ability to reproduce and may contribute to reproductive strategies of the population, any adolescent male greater than 50 mm CW and all adult males (of any size) were considered potentially sexually mature (Sainte-Marie et al., 1995) and used in the following exploratory analyses to identify influential factors affecting female reproductive metrics.

A suite of male-specific indices including average carapace width (CW_M) and chela height (CH) were determined at each site during 2017 surveys, with these males representing those potentially available to mate with primiparous females during the winter of 2018. An average index of time since terminal molt (TM) was estimated for all terminally molted males in 2017 using their shell conditions as a proxy: (a) soft or very soft = 0, (b) new hard = 1, (c) intermediate = 2, (d) old = 3, or (e) very old = 4. Since there could be a number of discrepancies associated with identifying shell conditions and the time since terminal molt, these values were meant to be used as a general index of time since terminal molt (i.e., relative age of the terminally molted population), rather than a precise estimation. Terminally molted individuals were identified using the methods described in Mullowney and Baker (2020a), whereby discrimination based on chela height to CW relationships is used (Baker et al., 2021). Finally, for males, the ratio of sexually competitive to less competitive crab (competitive index, CI) was identified for each site (as a percentage) using 2017 survey density data: competitive males / (competitive males + inferior males). Sexually competitive males were considered any terminally molted crab greater than 94 mm CW, while less competitive (inferior) males were all other sexually mature males.

Across-sex competition indices were also explored. Operational sex ratio (OSR) is defined as the ratio of sexually mature males to receptive females (Emlen and Oring, 1977), but as primiparous and multiparous females may not routinely compete for mates due to being receptive at different locations and times in the year, sex ratio calculations were restricted to primiparous females. Therefore, an annual, site-specific OSR_P was calculated as the average survey catch rates of primiparous females in 2018 in relation to the average survey catch rates of males during 2017 surveys: (Males / (Males + Primiparous females) x 100).

2.5. Data analyses

The relationship between SL and sperm count (N = 76) was

examined using a general linear model to ensure that SL was an accurate metric to identify the amount of sperm stored in a seminal receptacle (Sainte-Marie et al., 2002). The relationship between SL and sperm count was also examined to determine if there were significant differences between the sites.

Data exploration for important determinants of female reproductive output was performed following statistical and analytical protocols outlined in Zuur et al. (2010). Seminal receptacle load was examined in relation to CW_F, CH, CW_M, TM, OSR_P, CI, and the month of female sampling (to represent the time since molting). To avoid multi-collinearity, explanatory variables included in each model were reduced until all had variance inflation factors (VIF) less than 4 (Zuur et al., 2010). A generalized additive model (GAM) with a Tweedie distribution using the *mgcv* package (Wood, 2017) in R version 3.6.1 (R Core Team, 2020) was used to investigate the significant predictors of SL. A Tweedie distribution was chosen because SL had a continuous distribution that included zeros and it allows for flexible overdispersion.

Clutch weight was examined in relation to CW_F, AW_F, month sampled, and number of limbs missing in relation to the sampled female, as well as CW_M, CH, TM, OSR_P, and CI at the site. The number of limbs missing was used to represent the overall health of the sampled individual. Female SL was also considered as a potential predictor, but since sampling month was determined to significantly influence SLs (see Results and Discussion), sampled SLs were standardized to May values to account for varying sampling periods between sites. A GAM with a Gamma distribution (log link) was used to identify significant predictors of clutch weight. A Gamma distribution was chosen because clutch weight had a right-skewed, continuous distribution without zeros.

3. Results

A total of 268 primiparous females were sampled from 25 sets throughout the sites (Fig. 1, Table 1). The females ranged in size from 37.1 mm to 79.8 mm (median = 57.35 mm) CW (Fig. 2A). Females were overall largest in Fortune Bay with their lower interquartile range (IQR) limit for CW overlapping the upper IQR limits for CW of only Conception and Placentia Bays (Fig. 2A). Overall, approximately one-third (35%) of primiparous females had partial egg clutches, while the rest were considered full (Fig. 3). The primiparous females were deemed to be in generally good physical condition with only 23% of sampled females missing one or more limbs and a maximum of three missing limbs observed on three individuals (one in Notre Dame Bay, and two in Placentia Bay).

The six sites varied in descriptive characteristics of the males in 2017 and in OSR_P (Fig. 4). CW_M ranged from 61.9 mm in 3L200 to 85.4 mm in Conception Bay, and similarly CH ranged from 11.1 mm to 19.1 mm at these sites. The TM was lowest in 3L200 and highest in Conception Bay. Placentia Bay had the lowest CI, while 3L200 had the highest CI. Finally, the OSR_P was highest in 3L200 and lowest in Notre Dame Bay.

3.1. Seminal receptacle load

Overall, a strong, significant linear relationship was found between the subsamples of sperm counts and the SLs in all sites combined ($p < 0.001$, $R^2 = 0.91$, Eq. (1); Fig. 5). Although sperm counts and SLs varied between sites, there were no significant differences in regressions among the sites ($p > 0.05$). Therefore, consistent with other studies (e.g., Sainte-Marie et al., 2002), SL was found to be an appropriate index for sperm count following a linear transformation of the form:

$$SC = 438\ SL \tag{1}$$

Where, SC is sperm count (millions) and SL is seminal receptacle load (g).

Sampled (unstandardized) SLs ranged from undetectable to 0.7295 g, with an overall median of 0.0315 g (Fig. 2B). Fortune Bay had higher SLs (average 3.6x) than other sites. An empty seminal receptacle was observed in one female from each of Fortune Bay, Placentia Bay, and 3L200.

The best-fitting model predicting SL was a GAM with a Tweedie (tw) family (log link) distribution and significant predictors of CW_F ($p = 0.015$), Month ($p < 0.001$), TM ($p < 0.001$), and CI ($p < 0.001$) (42.4% deviance explained, Eq. (2), Table 2).

$$\begin{aligned} &SL_i \sim \text{Tweedie}(u_i, \varphi, P) \\ &E(SL_i) = u_i \quad var(SL_i) = \varphi u_i^P \\ &\log(u_i) = \beta_o + f_1(CW_{Fi}) + f_2(\text{Month}_i) + f_3(TM_i) + f_4(CI_i) \end{aligned} \tag{2}$$

where, SL_i is seminal receptacle load (g), the Tweedie distribution has three parameters: u_i is the mean, φ is the dispersion factor, and P is power (where $1 < P < 2$), β_o is the intercept, and the f_i's are smooth functions of the given covariates: CW_{Fi} is the carapace width (mm) of sampled female, Month_i is the month in which sampling took place (and represents an index of time since female mating), TM_i is the site-specific average index of time since terminal molt for males, and CI_i is the site-specific percentage of competitive males compared to less competitive males. The smooth functions were computed with thin-plate regression splines with a maximum basis dimension of 4 for Month_i and 6 for TM_i and CI_i, resulting from limited data availability.

Partial-effect plots showed that as CI and TM increased SL of primiparous females increased in linear fashion (Fig. 6). Similarly, as time (month) increased, a linear decline in SL occurred. A dome-shaped response occurred for the partial effect of CW_F on SL, with highest loads predicted at an intermediate size (60 mm CW_F) and lower SLs at small (30 mm CW_F) and large (80 mm CW_F) sizes. However, there was considerable variability at the tail-ends of the distribution, thus the true underlying relationship to CW remains somewhat unclear.

When SLs were standardized to the month of May using Eq. (2), a somewhat different picture of SLs between sites appeared, with Conception Bay having the highest median standardized SLs (0.259 g), followed by Notre Dame Bay (0.096 g), Fortune Bay (0.084 g), White Bay (0.067 g), Placentia Bay (0.060 g), and then 3L200 with the lowest standardized SLs (0.056 g) (Fig. 2B). The overall median May-standardized SL was 0.0813 g.

3.2. Clutch weight

Clutch weight ranged from 0.1599 g (3L200) to 13.1722 g (Fortune Bay), with an average of 4.6048 g across all sites (Fig. 2C). Fortune Bay had the largest average clutch weight, while clutch weights at 3L200 were substantially smaller compared to other sites.

The best-fitting model predicting clutch weight was a GAM with a Gamma family (log link) distribution and featuring CW_F ($p < 0.001$), CW_M ($p < 0.001$), and TM ($p = 0.004$) as significant predictors of clutch weight (79.3% deviance explained, Eq. (3), Table 3)

$$\begin{aligned} &Y_i \sim \text{Gamma}(u_i, r) \\ &E(Y_i) = u_i \quad var(Y_i) = u_i^2/r \\ &\log(u_i) = \beta_o + f_1(CW_{Fi}) + f_2(CW_{Mi}) + f_3(TM_i) \end{aligned} \tag{3}$$

Table 1
Summary of samples collected in 2018 off Newfoundland. The number of sets, number of primiparous female snow crab sampled, and months of collections at each site are provided.

Location	# of sets	# of females sampled	Month
Placentia Bay (PB)	1	40	June
Fortune Bay (FB)	4	53	May–June
White Bay (WB)	8	54	September
Notre Dame Bay (NDB)	7	40	September
Conception Bay (CB)	4	49	October
3L200	1	32	Late–August
Total	25	268	

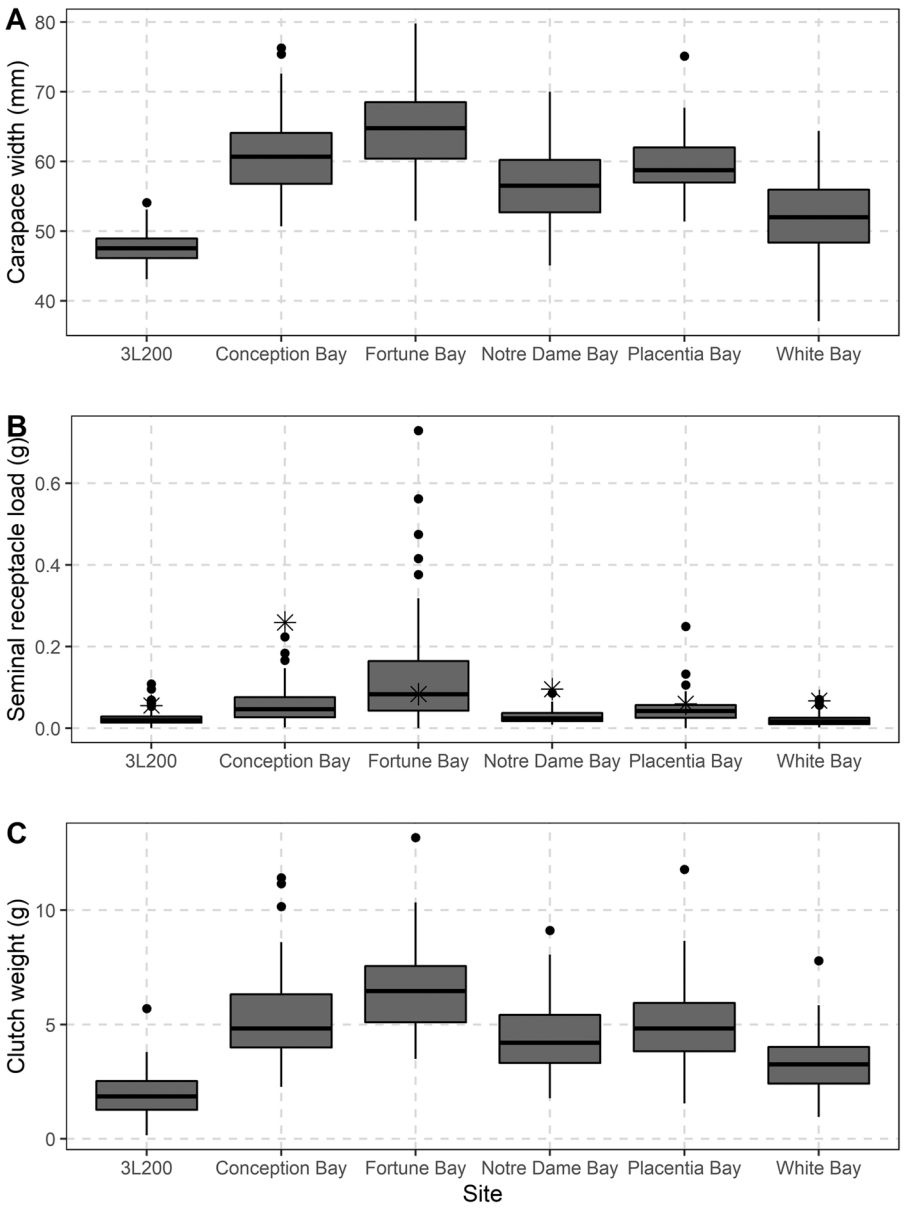


Fig. 2. Boxplots of (A) carapace width (mm), (B) seminal receptacle load (g), and (C) clutch weight (g) of primiparous female snow crab sampled at six sites off Newfoundland in 2018. Box encompasses 75th and 25th percentiles determining the Interquartile Range (IQR), with whiskers extending outside of the box to a distance of 1.5 × IQR, outlier values are black circles, and medians are solid black lines. Asterisks in plot B represent median May-standardized seminal receptacle loads.

where, Y_i is clutch weight (g), β_o is the intercept, the f_i's are smooth functions of the given covariates: CW_{Fi} is the carapace width (mm) of sampled females, CW_{Mi} is the site-specific average carapace width of males, and TM_i is the site-specific average index of time since terminal molt for males. The smooth functions were computed with thin-plate regression splines with a maximum basis dimension of 6 for CW_{Mi} and TM_i, resulting from limited data availability. Partial-effects plots illustrate that clutch weight increases as both CW_F and CW_M increase (Fig. 7). As TM increased the clutch weight decreased. Standardized SL was not a significant predictor of clutch weight.

4. Discussion

4.1. Seminal receptacle load

With no control site and no temporal coverage of SL available from NL, interpretation of results is restricted to a qualitative assessment of SL of primiparous females within the NL snow crab stock range, and definitive conclusions on the reproductive health status of the stock are difficult to make. May-standardized NL snow crab SLs were compared to samples collected in late-April to mid-May over the course of three-decades in the Northern Gulf of the St. Lawrence (nGSL) (Sainte-Marie et al., 2002, 2008; B. Sainte-Marie, unpubl. data). In comparison to the nGSL, a population within the same genetic stock range as NL snow crab (Puebla et al., 2008), SLs in NL snow crab are low relative to historically defined benchmarks (Sainte-Marie and Sainte-Marie, 1998). The medians, averages, and upper 75th percentiles of May-standardized SLs at all our study sites, with the exception of Conception Bay, were below the three-decade average levels for primiparous females from the nGSL (overall average of annual medians: 0.1550 g; B. Sainte-Marie, unpubl. data) and similar only to the very lowest values of the time series. In contrast to some concerning signals arising from the comparison to the adjacent nGSL snow crab, our sampled SLs at NL study sites rendered similar values to those reported in a short-term study in the Eastern Bering Sea (EBS) of Alaska during 2005–2008 (Slater et al., 2010).

Specific reasons for differences in SLs of snow crab from the nGSL and EBS are unknown. However, the EBS surveys from which primiparous females were collected were conducted from the end of May to the end of July, so it is not unreasonable to suppose that mean collection date lagged nGSL sampling by about 1–2 months. It is possible that the

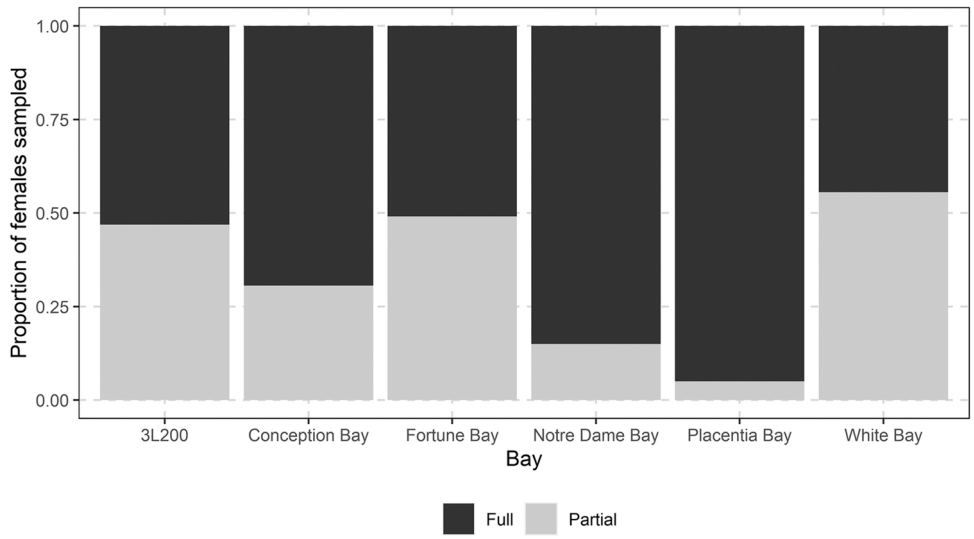


Fig. 3. The proportion of primiparous females sampled that were identified as having partial (light gray) and full (black) egg clutches within each study site collected in 2018 off Newfoundland.

differences in SL between EBS and nGSL may, to some extent, reflect a difference in sampling time. In addition, we speculate ecosystem scales could be an important factor and that the expansive NL marine shelf ecosystem is more akin to the EBS than to the nGSL ecosystem. Distance traveled both during downslope ontogenetic movements and upslope seasonal male breeding migrations are known to be much larger in NL and the EBS than in the nGSL (Ernst et al., 2005; Parada et al., 2010; Mullowney et al., 2018a). It is plausible that greater distances between breeding sites or more prolonged separation among breeding groups of crab for much of the year could reduce mate encounter rates in the larger ecosystems. Furthermore, primiparous females spatiotemporally overlap more commonly with adolescent and small adult males in expansive shelf systems (Parada et al., 2010), compared to closer associations with small and large adult males in narrow shelf systems (Lovrich et al., 1995). These distributional dynamics could affect average SL, with adolescent and small adult males having smaller sperm reserves (Sainte-Marie et al., 1995) and smaller ejaculates compared to large adult males (Sainte-Marie and Lovrich, 1994).

Following the concept of higher potential for sperm limitation in more expansive habitats, study areas that occur within or form a continuum of the offshore, such as 3L200, Notre Dame Bay, and Placentia Bay, might be expected to have primiparous females featuring lower SLs than those in more spatially confined inshore bays such as Fortune Bay and Conception Bay. Our results were suggestive of this possibility. However, the relatively low standardized SLs in White Bay, a deep fjord ecosystem, would appear to suggest that ecosystem scale alone does not sufficiently explain differences in observations among sites and reinforce the possibility that SLs in the NL study sites are generally below levels where they would be if the population was reproductively healthy.

Our observations on SLs in NL snow crab in comparison to the nGSL suggest that there is likely less reproductive resilience to fishing within many areas of the NL marine ecosystem. Although many females in our study did indeed have full clutches of eggs, the overall lower average levels of sperm stored in the seminal receptacles highlight reduced potential to autonomously fertilize one or more other egg clutches (i.e., as multiparous females) in the NL populations. This is important because despite nGSL and NL snow crab being managed and assessed by different regional institutes, DFO strives to manage all snow crab populations in Atlantic Canada under similar strategies across regions (FRCC (Fisheries Resource Conservation Council), 2005).

A potential strength of a male-only fisheries management strategy is constancy in reproductive resilience to fishing (Ennis et al., 1990; Kruse, 1993). Constancy of this strategy revolves around ensuring adequate sperm supply in both the present (i.e., egg clutches) and the future (i.e., via storage of surplus sperm in female seminal receptacles). Multiparous females drawing from inadequate sperm reserves are not capable of fully fertilizing the next egg clutch and need to mate again in the following breeding season (Yamasaki et al., 1994; Sainte-Marie et al., 2002; Webb et al., 2016). Although we cannot concretely conclude that sperm limitation is occurring within the NL snow crab resource to an extent strong enough to diminish stock-level reproductive capacity, the evidence enables a conclusion that insemination rates of females were generally low in 2018, thus the constancy of reproductive resilience under the male-only fishing strategy may not be assured.

Following Rondeau and Sainte-Marie (2001), which uses size-fecundity relationship of multiparous female snow crab (Sainte-Marie, 1993) and the approximation of 70 sperm cells needed to fertilize a single oocyte (Sainte-Marie and Lovrich, 1994), along with Eq. (1) and sampled SLs standardized to the latest month within our data range (October) based on Eq. (2), we estimate that at least 7% (N = 18) of the sampled NL females did not have enough stored sperm to fully fertilize a subsequent full egg clutch without being re-mated. More concerning, 12 of these 18 females had partial clutches during sampling and therefore appeared to have lacked enough sperm to fertilize their first clutch. Notably, half (9/18) of the females were from Fortune Bay. In addition, females delay extruding eggs when there are fewer than 1.9×10^5 sperm cells per seminal receptacle to allow for the opportunity of mating with another male (Sainte-Marie and Lovrich, 1994). Eventually, if more sperm cells are not obtained, the female will either resorb the oocytes or extrude an unfertilized or incompletely fertilized egg clutch (Sainte-Marie et al., 2010). Three females sampled for this study would not have had the minimum sperm counts needed to meet this threshold. Our qualitative assessment suggests the reproductive buffering capacity of this stock to persistent high fishing pressure is relatively low.

With respect to factors affecting SLs, we found significant effects of the male competition index, average time since male terminal molt, month, and female CW. The process surrounding the benefits of a high ratio of competitive males in the population would appear straightforward. As the proportion of the large, terminally molted males increases, intensity of male sexual competition, and frequency of successful mating events increase and average SL delivered by males increases. Although this conforms with the general theories of sperm economy and male snow crab behavior (Sainte-Marie et al., 2008), to our knowledge it has not previously been observed in a wild population of snow crab.

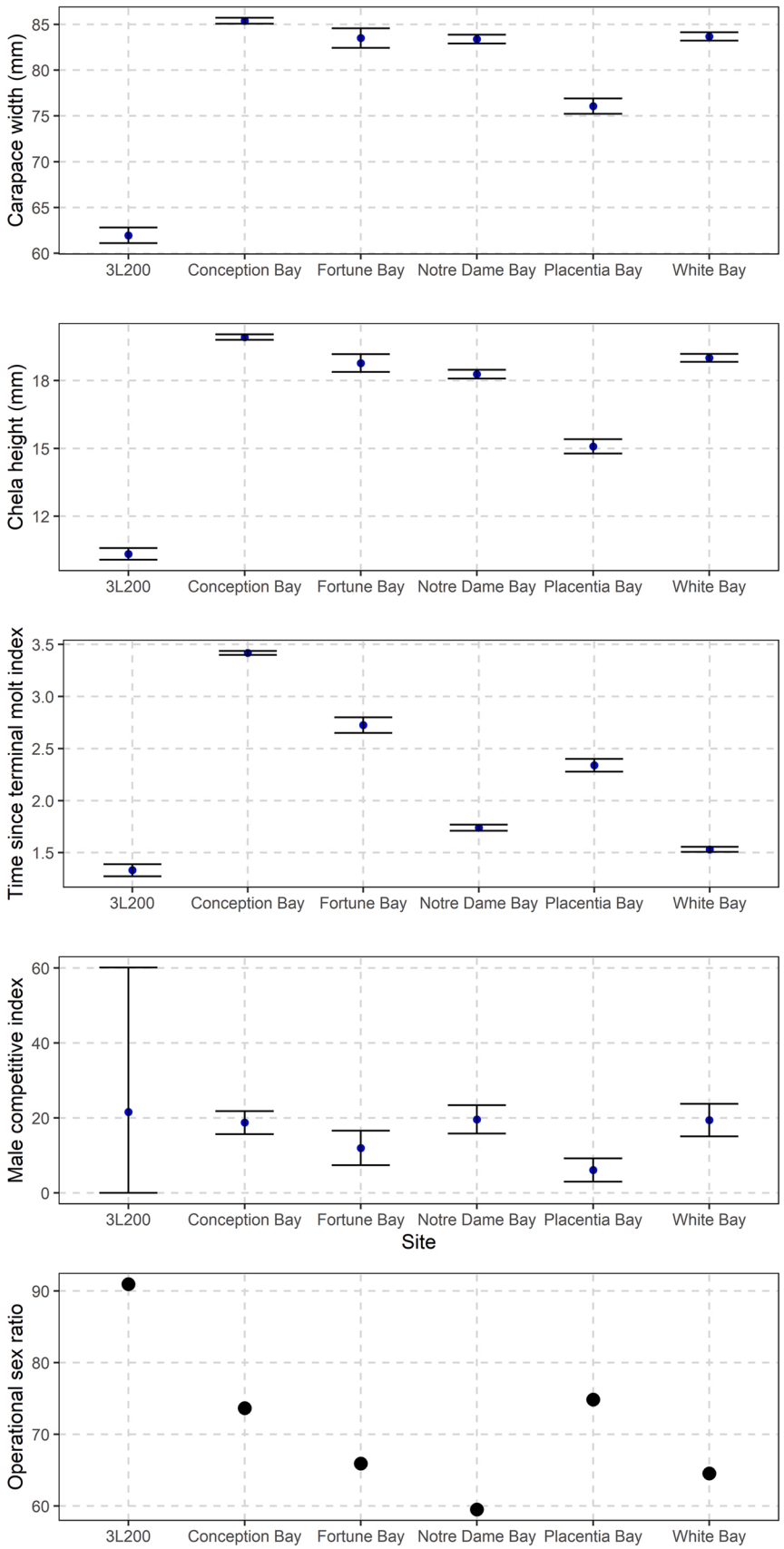


Fig. 4. Average and 95% confidence interval (where possible) of male carapace width (mm), chela height (mm), time since male terminal molt index, competitive index, and population operational sex ratio [males / (males + primiparous females)] x 100 at six sites off Newfoundland.

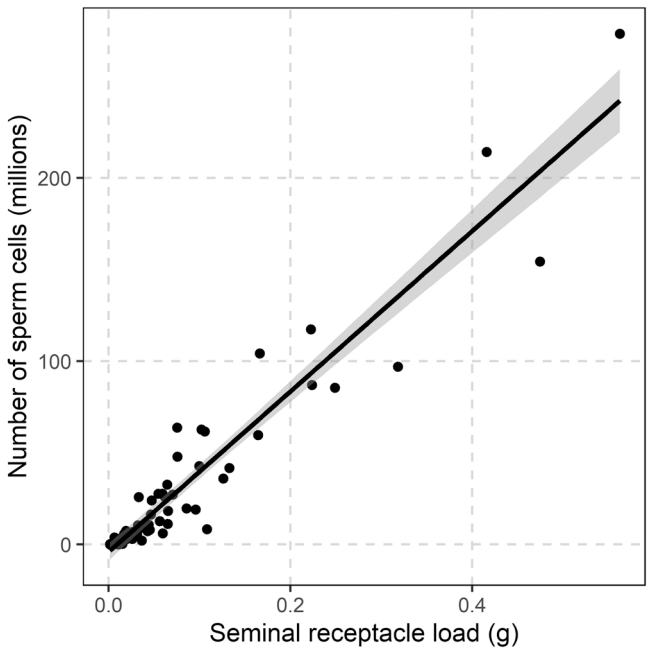


Fig. 5. Number of sperm cells (millions) in relation to seminal receptacle load (g) from samples of primiparous female snow crab collected in 2018 at six sites off Newfoundland.

Table 2
Model summary of generalized additive model with exploratory variables of seminal receptacle loads in female snow crab at six sites off Newfoundland in 2018.

Explanatory variable	Estimated degrees of freedom	Reference degrees of freedom	Parameter estimate	Standard error	p-value
Female carapace width (CW_F)	2.313	9			0.015
Month	1.102	3			< 0.001
Index of age since male terminal molt (TM)	0.984	5			< 0.001
Competitive index (CI)	0.930	5			< 0.001
Intercept			-3.1038	0.0483	< 0.001

The linear and positive predictive relationship between the male average time since terminal molt and SL up to a shell condition index of 3.5 is somewhat unexpected. Although our index does not directly conform to years, an index of 2 would closely align with what we would commonly assess as a crab in an 'intermediate' stage condition which is considered to be in peak physical condition (Sainte-Marie et al., 1999; Fonseca et al., 2008). It takes up to a year for shells to become fully hard and filled with meat following a molt (Mullowney et al., 2020), and it is reasonable to suggest that the next 1–2 years of the crab's life following this phase would represent peak condition before the shell begins to deteriorate and the male becomes senescent (Sainte-Marie et al., 1999). Therefore, we expected a peak near the central part of a 0–4 stage shell condition index. However, our results are consistent with other studies that found older shell males typically have larger sperm reserves at size than younger shell males (Sainte-Marie et al., 1995). In an experiment reported by Sainte-Marie et al. (2008), primiparous females had 39% larger SLs after mating and spawning with adult old-shell males, compared to similarly-sized intermediate-shelled adult males. This conforms with our results that suggest ageing adult males may pass more

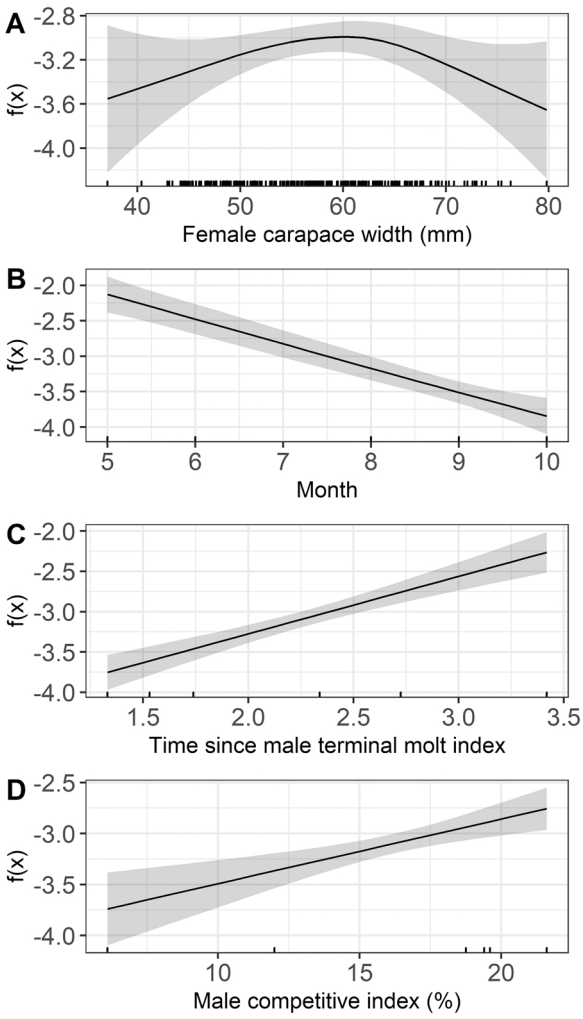


Fig. 6. Log-scale partial-effects plot of (A) sampled female carapace width (mm), (B) month when sampling occurred, (C) average index of age since terminal molt (males), and (D) percentage of superior males compared to all potentially sexually active males (%) on seminal receptacle load (g) in female snow crab off Newfoundland. The rug marks along the x-axis represent observed values.

Table 3
Model summary of generalized additive model with exploratory variables of egg clutch weight of female snow crab at six sites off Newfoundland in 2018.

Explanatory variable	Estimated degrees of freedom	Reference degrees of freedom	Parameter estimate	Standard error	p-value
Female carapace width (CW_F)	3.025	9			< 0.001
Average male carapace width (CW_M)	1.017	5			< 0.001
Index of average age since male terminal molt (TM)	0.849	4			0.004
Intercept			1.4147	0.1273	< 0.001

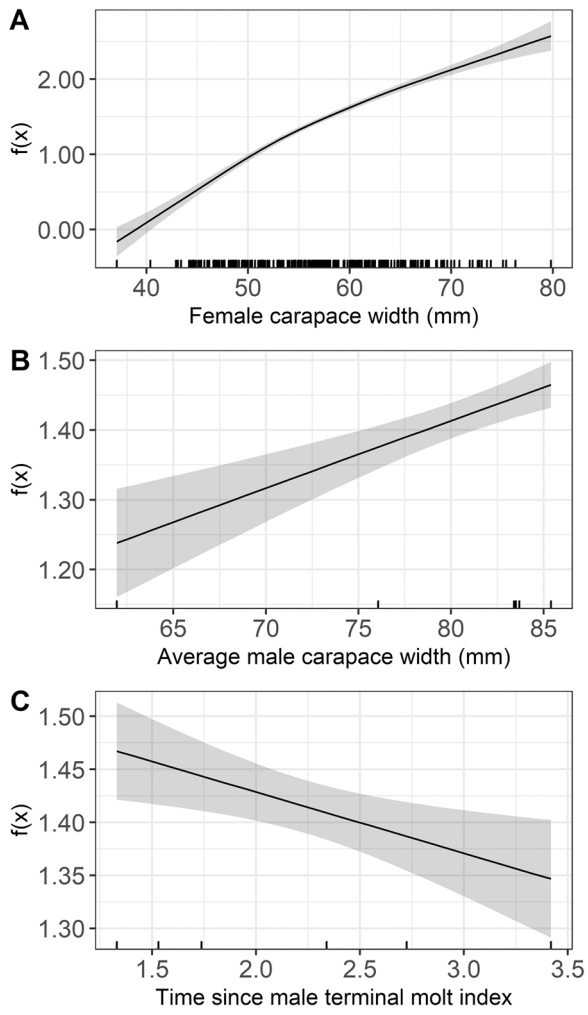


Fig. 7. Log-scale partial-effects plot of (A) sampled female carapace width (mm), (B) average male carapace width of potentially sexually responsive males (mm), and (C) average index of age since terminal molt (males) on clutch weight (g) in female snow crab off Newfoundland. The rug marks along the x-axis represent observed values.

sperm to females than young adult males.

We found a significant negative relationship between female time since molting/mating (represented here by month) and SL. Although we cannot rule out that month may also represent variance associated with unmeasured site-specific characteristics, the loss of SL after mating is a recognized phenomenon in snow crab (Sainte-Marie, 1993). The gravimetric loss of SL over time in snow crab was documented through repeated sampling throughout a year and was ostensibly due to attrition of amorphous matter (i.e. seminal fluids), which represents the largest part of ejaculates (Sainte-Marie and Sainte-Marie, 1998). However, it is also likely accompanied by some loss of sperm cells (see general discussion in Elner and Beninger, 1995) although this has not been quantified to date. Indeed, the lack of difference between months/sites in the relationship of SC to SL in our study can suggest proportional decline of sperm cell number and SL. In populations of the subtropical blue crab *Callinectes sapidus*, females may lose 50–95% of their stored sperm cells in the first 2–3 months after mating and before spawning possibly due to erosion of the external sperm plug (Wolcott et al., 2005; Rains et al., 2018; Ogburn et al., 2019). The extent of female sperm losses may likely vary from one species (mating system) to another, and perhaps also within species as a function of the occurrence and prevalence of bacterial infections (Benhalima and Moriyasu, 2001) or other pathologies. Nevertheless, 6–11 months or 18–19 months would have elapsed between the time we sampled primiparous females and the time these females would have spawned again under scenarios of temperature-dependent annual or biennial reproductive cycle (Moriyasu and Lanteigne, 1998), respectively, suggesting the likely possibility that our sperm counts overestimate the potential for autonomous fertilization.

Finally, we found a dome-shaped partial effect of CW_F on SL; highest SLs were predicted in mid-sized females of approximately 60 mm CW_F. Past research has found that males do not adjust their size of ejaculation to the size of the female (Adams and Paul, 1983; Sainte-Marie and Lovrich, 1994), but since females can be polyandrous it is possible that the largest females could have less sperm reserves because they were mated by too few males for their overall size. Largest females ultimately require largest SLs to fully fertilize larger egg clutches, and larger females tend to molt before smaller females during the primiparous breeding season, which should theoretically help them acquire mates before males become sperm-depleted (Rondeau and Sainte-Marie, 2001). Nevertheless, there is a possibility that low SLs in the largest females are reflective of sperm limitation occurring in the population. Post-hoc exploratory analyses further strengthen this interpretation. When Fortune Bay females were removed from the SL model, the relationship between SLs and CW_F became overall positive, but non-significant ($p = 0.275$), even though CW_F values exceeding 70 mm were still present. This indicates that the tail-end of the dome-shaped curve is likely a result of the largest females in Fortune Bay having SLs smaller than expected based on their size and suggests sperm limitation is an issue at this site.

4.2. Clutches

A relatively large proportion of the females had partial clutches, with the highest proportions of partial clutches noted in White Bay, Fortune Bay, and 3L200. Given the flexible reproductive behavior of male and female snow crab (Sainte-Marie et al., 2008), under healthy conditions most females would be expected to have full clutches (Mullowney et al., 2018b). Post-hoc analyses to model drivers of partial versus full clutches were unsuccessful. However, there appeared to be a general, albeit non-significant ($p = 0.092$), relationship with CW_F, where the odds of having a full clutch peaked near 55 mm CW and larger females had reduced odds of a full clutch. Nevertheless, the ultimate causes of the frequent partial clutches observed in our study remain unknown but fit within the general themes of the results in raising concerns for the overall reproductive health at these sites.

We found that clutch weight had a positive relationship with CW_F. Given the strong collinearity between CW_F and AW_F ($R^2 = 0.94$), they could not both be included in a single model. Nevertheless, both metrics influence a female's clutch weight. Simply, larger females are able to produce and carry larger clutches. While CW_F directly influences maximum ovary size (i.e., the body cavity volume) (Hines, 1982; Sainte-Marie and Lovrich, 1994), AW_F, in theory, determines the maximum number of eggs that can be brooded to full development (Hartnoll, 1974; Alunno-Bruscia and Sainte-Marie, 1998). Although a female's clutch size is proportional to its overall size, a male's sperm allocation is not proportional to female size (Adams and Paul, 1983; Sainte-Marie and Lovrich, 1994). As described above, this could result in areas with larger females being more prone to sperm limitation causing egg losses and smaller clutches. Fortune Bay had the largest females and associated clutch weights among the sampled sites, but despite also having the largest sampled SLs (and third highest month-standardized sperm loads), a high percentage (49%, $N = 26$) of the sampled females in Fortune Bay had partial clutches. Although these observations are suggestive of sperm limitation occurring in Fortune Bay, it should be cautioned that the high proportion of partial clutches could also reflect other processes. For example, although Fortune Bay was the earliest sampled site, larger more fecund females are known to have proportionately greater egg losses throughout the breeding season, resulting

from a variety of factors including predation or abrasion of live eggs (Elner and Beninger, 1992).

Clutch weight was also significantly related to site-specific characteristics of males. CW_M positively affected clutch weight which aligns with the notion that large males are overall superior breeders (Sainte-Marie et al., 2008). Prior to the snow crab fishery being fully expanded throughout NL, Taylor (1996) found that both fecundity and clutch weight were lower in study sites with commercial exploitation, compared to unexploited areas off Newfoundland, suggesting the removal of prime male breeders by the fishery was an important factor. The present results reinforce the efficacy of a strategy of maintaining a relatively high abundance of large males in the population as a central tenet of fisheries management to help maintain stock reproductive capacity.

Finally, our models illustrated that clutch weight had a significant negative relationship with male TM. Specific reasons for the direction of this relationship are unknown. However, as per our previous descriptions on the male time-since terminal molt, a dome-shaped or positive effect in this index was anticipated.

4.3. Other considerations

Our study produced some results that do not align with previous studies. We found a lack of relationship between May-standardized SL and clutch weight. In the nGSL, there is a positive relationship between these two characteristics and SL is considered the most proximate factor in fertilization success (Sainte-Marie et al., 2008) even though SL measured post-spawn may not always proportionately represent what was available pre-spawn. Our lack of relationship remains perplexing but leaves open the idea that perhaps there are other important factors related to the SLs (e.g., sperm quality) that have an impact on fertilization success and embryo survival (Aquino et al., 2021), in comparison to just the size of the SL. In addition, the best model for explaining SLs in this study did not include OSR_P, an index that would seem plausibly related to reproductive integrity in the stock (e.g., Sainte-Marie et al., 2002). This could reflect the lack of temporal (interannual) variability in our study, where recruitment pulses that cause large changes in OSR_P through time (Sainte-Marie et al., 2002, 2008) were not represented in our analyses.

The male-only fishery of snow crab off NL directly influences three of the significant predictors of SL and clutch weight (CI, TM, and CW_M) identified in our study. The fishery targets the most competitive proportion of the male population (hard-shelled, large individuals, most of which are terminally molted) and high exploitation pressure can greatly reduce the average size of males in the population (Mullowney and Baker, 2020a), as well as the proportion of competitive males. Such reductions in-turn are likely to reduce both SLs and clutch weights. The observations from the persistently poor state study site (Baker et al., 2021), Fortune Bay, are particularly important in this regard as they demonstrate potential long-term outcomes. Resource depletion in recent years has also greatly increased the level of discards (both legal soft-shelled and under-sized hard-shelled males) observed in the NL fishery (Baker et al., 2021; Mullowney et al., 2021). The mortality and injury rate of discarded soft-shelled males is unknown, but high discard rates could result in significant numbers of individuals never becoming competitive males (Mullowney et al., 2021). Strong exploitation pressure will result in fewer individuals surviving one or two years beyond terminal molt. For example, in NAFO Divisions 2HJ off NL (north of sites sampled in this study), snow crab are being fished so heavily that very few hard-shelled, exploitable crab remain after the fishery is complete and the fishery is leaning toward a recruitment fishery (Baker et al., 2021). Virtually no males would reach a time since terminal molt index of 2–2.5.

5. Conclusions

Although a male-only fishery, the snow crab fishery off NL can strongly influence female reproductive health. The results of our research illustrate the importance of conservative exploitation rates that keep an adequate proportion of large, hard-shelled males in the population. Such measures are deemed necessary to ensure that the low sperm reserves observed in this study do not hamper reproductive potential of the population.

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CRediT authorship contribution statement

Krista Baker: Conceptualization, Methodology, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration. **Darrell Mullowney**: Conceptualization, Writing – original draft, Writing – review & editing. **Bernard Sainte-Marie**: Methodology, Validation, Investigation, Writing – review & editing, Mentoring.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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